

Solution Manual

Analysis and Design of Elementary MOS Amplifier Stages

**- First Edition -
Chapter 4**

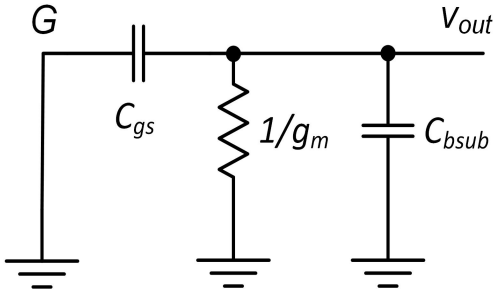
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`restart : with(Units[Standard]) : with(plots) : interface(imaginaryunit=j) : read "Parameters.mpl" :`

P4.1

$L := 2 \llbracket \mu m \rrbracket : V_{OUT} := 1.5 \llbracket V \rrbracket : I_B := 200 \llbracket \mu A \rrbracket : X_1 := 5 \llbracket \mu m \rrbracket : X_2 := 3 \llbracket \mu m \rrbracket :$

(a) The small-signal model is shown below. C_{gs} and C_{sb} are not included, since they are shorted and therefore irrelevant for the circuit's operation. g_{mb} is also irrelevant since $v_{bs} = 0$



(b) We begin by determining the width of the transistor using the given information.

$$I_B = \frac{1}{2} \mu_p C_{ox} \cdot \frac{W}{L} (V_{SG} + V_{T0p})^2 :$$

$$W := \frac{I_B}{\frac{1}{2} \mu_p C_{ox} \cdot \frac{1}{L} (V_{OUT} + V_{T0p})^2} = 0.000 \llbracket m \rrbracket \xrightarrow{\text{replace units}} 32.000 \llbracket \mu m \rrbracket$$

We can now compute C_{gs} and C_{bsub} (using the method of Example 4-1) as follows.

$$C_{gs} := \frac{2}{3} \cdot W \cdot L \cdot C_{ox} + W \cdot C_{ov} = 1.141 \cdot 10^{-13} \llbracket F \rrbracket$$

$$A_{well} := (W + 2 \cdot X_1) \cdot (L + 2 \cdot L_{diff} + 2 \cdot X_1 + X_2) = 8.820 \cdot 10^{-10} \llbracket m^2 \rrbracket$$

$$C_{bsub} := \frac{C_{Jwell} \cdot A_{well}}{\left(1 + \frac{V_{OUT}}{PB}\right)^{MJ}} = 2.746 \cdot 10^{-14} \llbracket F \rrbracket$$

The ratio of well-substrate and gate-source capacitance is

$$\frac{C_{bsub}}{C_{gs}} = 0.241$$

4.2

$$V_{DD} := 5 \text{ [V]} : V_B := 2.5 \text{ [V]} : I_B := 200 \text{ [uA]} : R_D := 3 \text{ [k}\Omega\text{]} : W := 100 \text{ [um]} : L := 1 \text{ [um]} :$$

Note: The problem statement in the first edition contains an error (see errata list). The problem should be worked for $I_B = 200\mu\text{A}$ (instead of $400\mu\text{A}$), as done here.

$$V_{OUT} := V_{DD} - 1 \cdot I_B \cdot R_D = 4.400 \text{ [V]}$$

$$V_{OV} := \sqrt{\frac{2 \cdot I_B}{\mu n C_{ox} \cdot \frac{W}{L}}} = 0.283 \text{ [V]}$$

$$V_S := V_B - V_{T0n} - V_{OV} = 1.717 \text{ [V]}$$

$$V_{DS} := V_{OUT} - V_S = 2.683 \text{ [V]}$$

4.3

$$V_{DD} := 2.5 \llbracket V \rrbracket : V_{SS} := -2.5 \llbracket V \rrbracket : P_{DIS} := 1 \llbracket mW \rrbracket : R_D := 5 \llbracket k\Omega \rrbracket : R_S := 5 \llbracket k\Omega \rrbracket : L := 2 \llbracket um \rrbracket : A_i := 0.8 :$$

Note: The problem statement in the first edition contains several errors (see errata list). The problem is worked here according to the corrections stated in the errata list.

(a) From the given power dissipation we can calculate I_{SUP} .

$$I_{SUP} := \frac{P_{DIS}}{V_{DD} - V_{SS}} = 0.000 \llbracket A \rrbracket \xrightarrow{\text{replace units}} 200.000 \llbracket uA \rrbracket$$

The output resistance of the CG stage is approximately R_S times the intrinsic gain of the transistor. It will therefore be orders of magnitude larger than R_D , even if channel-length modulation was considered. We can therefore safely neglect the output resistance in the current gain calculation, which is then simply given by the current divider at the source of the transistor:

$$A_i = \frac{i_{out}}{i_s} = \frac{g_m}{g_m + \frac{1}{R_S}}$$

$$g_m := \frac{1}{R_S} \cdot \left(\frac{1}{A_i} - 1 \right)^{-1} = 0.001 \llbracket S \rrbracket \xrightarrow{\text{replace units}} 0.800 \llbracket mS \rrbracket$$

$$W := \frac{g_m^2}{2 \cdot I_{SUP} \cdot \mu_p C_{ox} \cdot \frac{1}{L}} = 0.000 \llbracket m \rrbracket \xrightarrow{\text{replace units}} 128.000 \llbracket um \rrbracket$$

(b) With the bulk connected to the positive supply, the current gain is now

$$A_i = \frac{i_{out}}{i_s} = \frac{g'_m}{g'_m + \frac{1}{R_S}}$$

To calculate g'_m , we need to know V_{BS} . Similar to the way we approached this in Example 4-3, we can obtain a good and simple guess of this voltage by neglecting the backgate effect in the bias point calculation.

$$V_{OV} := \frac{2 \cdot I_{SUP}}{g_m} = 0.500 \llbracket V \rrbracket$$

$$V_{BS} := V_{DD} - (-V_{T0p} + V_{OV}) = 1.500 \llbracket V \rrbracket$$

$$g_{mb} := g_m \cdot \frac{\gamma_n}{2\sqrt{2 \cdot \phi_f + V_{BS}}} = 0.000 \llbracket S \rrbracket \xrightarrow{\text{replace units}} 0.158 \llbracket mS \rrbracket$$

$$g'_m := g_m + g_{mb} = 0.001 \llbracket S \rrbracket \xrightarrow{\text{replace units}} 0.958 \llbracket mS \rrbracket$$

$$A_i := \frac{g'_m}{g'_m + \frac{1}{R_S}} = 0.827$$

If we iterate to obtain a better estimate of the threshold voltage, we obtain the following.

$$V_{Tp} := V_{T0p} - \gamma_p \cdot \left(\sqrt{2 \cdot \phi_f + V_{BS}} - \sqrt{2 \cdot \phi_f} \right) = -0.873 \text{ [V]}$$

$$V_{BS} := V_{DD} - (-V_{Tp} + V_{OV}) = 1.127 \text{ [V]}$$

$$g_{mb} := g_m \cdot \frac{\gamma_n}{2\sqrt{2 \cdot \phi_f + V_{BS}}} = 0.000 \text{ [S]} \xrightarrow{\text{replace units}} 0.173 \text{ [mS]}$$

$$g'_m := g_m + g_{mb} = 0.001 \text{ [S]} \xrightarrow{\text{replace units}} 0.973 \text{ [mS]}$$

$$A_i := \frac{g'_m}{g'_m + \frac{1}{R_S}} = 0.829$$

From here, we see that the iteration is not worth the effort. The noteworthy conclusion from this problem is the current gain increases when the bulk is connected to the supply instead of the source.

4.4

$$I_B := 400 \text{ [}\mu\text{A]} : R_D := 3 \text{ [k}\Omega\text{]} : W := 100 \text{ [}\mu\text{m]} : L := 3 \text{ [}\mu\text{m]} : V_{DD} := 5 \text{ [V]} : V_B := 2.5 \text{ [V]} :$$

First we compute the bias point parameters, following the approach of Example 4-1.

$$V_{OUT} := 1 \cdot V_{DD} - I_B \cdot R_D = 3.800 \text{ [V]}$$

$$V_{OV} := \sqrt{\frac{2 \cdot I_B}{\mu_n C_{ox} \cdot \frac{W}{L}}} = 0.693 \text{ [V]}$$

$$V_{SB} := V_B - V_{T0n} - V_{OV} = 1.307 \text{ [V]}$$

$$V_{Tn} := V_{T0n} + \gamma_n \cdot \left(\sqrt{2 \cdot \phi_f + V_{SB}} - \sqrt{2 \cdot \phi_f} \right) = 0.834 \text{ [V]}$$

$$V_{SB} := V_B - V_{Tn} - V_{OV} = 0.973 \text{ [V]}$$

This estimate is good enough to move on with the calculations. Below we run a few more iterations to get the exact value, though.

$$V_{Tn} := V_{T0n} + \gamma_n \cdot \left(\sqrt{2 \cdot \phi_f + V_{SB}} - \sqrt{2 \cdot \phi_f} \right) = 0.762 \text{ [V]}$$

$$V_{SB} := V_B - V_{Tn} - V_{OV} = 1.045 \text{ [V]}$$

$$V_{Tn} := V_{T0n} + \gamma_n \cdot \left(\sqrt{2 \cdot \phi_f + V_{SB}} - \sqrt{2 \cdot \phi_f} \right) = 0.778 \text{ [V]}$$

$$V_{SB} := V_B - V_{Tn} - V_{OV} = 1.029 \text{ [V]}$$

$$V_{Tn} := V_{T0n} + \gamma_n \cdot \left(\sqrt{2 \cdot \phi_f + V_{SB}} - \sqrt{2 \cdot \phi_f} \right) = 0.775 \text{ [V]}$$

$$V_{SB} := V_B - V_{Tn} - V_{OV} = 1.032 \text{ [V]}$$

$$V_{Tn} := V_{T0n} + \gamma_n \cdot \left(\sqrt{2 \cdot \phi_f + V_{SB}} - \sqrt{2 \cdot \phi_f} \right) = 0.776 \text{ [V]}$$

$$V_{SB} := V_B - V_{Tn} - V_{OV} = 1.032 \text{ [V]}$$

$$V_{Tn} := V_{T0n} + \gamma_n \cdot \left(\sqrt{2 \cdot \phi_f + V_{SB}} - \sqrt{2 \cdot \phi_f} \right) = 0.775 \text{ [V]}$$

$$V_{SB} := V_B - V_{Tn} - V_{OV} = 1.032 \text{ [V]}$$

We can now compute all of the required small-signal parameters.

$$g_m := \frac{2 \cdot I_B}{V_{OV}} = 0.001 \text{ [S]} \xrightarrow{\text{replace units}} 1.155 \text{ [mS]}$$

$$g_{mb} := g_m \cdot \frac{\gamma_n}{2 \sqrt{2 \cdot \phi_f + V_{SB}}} = 0.000 \text{ [S]} \xrightarrow{\text{replace units}} 0.256 \text{ [mS]}$$

$$g'_m := g_m + g_{mb} = 0.001 \text{ [S]} \xrightarrow{\text{replace units}} 1.411 \text{ [mS]}$$

$$C_{gs} := \frac{2}{3} \cdot W \cdot L \cdot C_{ox} + W \cdot C_{ov} = 5.100 \cdot 10^{-13} \text{ [F]}$$

$$C_{gd} := W \cdot C_{ov} = 5.000 \cdot 10^{-14} \text{ [F]}$$

$$AS := 1 \cdot W \cdot L_{diff} = 3.000 \cdot 10^{-10} \text{ [m}^2\text{]} \xrightarrow{\text{replace units}} 300.000 \text{ [}\mu\text{m}^2\text{]}$$

$$PS := 2 \cdot L_{diff} + W = 106 \text{ } [\mu m]$$

$$C_{sb} := \frac{C_{Jn} \cdot AS}{\left(1 + \frac{V_{SB}}{PB}\right)^{MJ}} + \frac{C_{JSWn} \cdot PS}{\left(1 + \frac{V_{SB}}{PB}\right)^{MJSW}} = 6.225 \cdot 10^{-14} \text{ } [F]$$

$$C_{db} := \frac{C_{Jn} \cdot AS}{\left(1 + \frac{V_{OUT}}{PB}\right)^{MJ}} + \frac{C_{JSWn} \cdot PS}{\left(1 + \frac{V_{OUT}}{PB}\right)^{MJSW}} = 4.441 \cdot 10^{-14} \text{ } [F]$$

$$C_s := C_{gs} + C_{sb} = 5.722 \cdot 10^{-13} \text{ } [F]$$

$$C_d := C_{gd} + C_{db} = 9.441 \cdot 10^{-14} \text{ } [F]$$

$$R_{so} := \frac{1}{g'_m} = 708.894 \text{ } [\Omega]$$

$$\tau_{so} := R_{so} \cdot C_s = 4.057 \cdot 10^{-10} \text{ } [s]$$

$$\tau_{do} := R_D \cdot C_d = 2.832 \cdot 10^{-10} \text{ } [s]$$

$$f_{3dB} := \frac{1}{2 \cdot 3.142} \cdot \frac{1}{\tau_{so} + \tau_{do}} = 2.310 \cdot 10^8 \left[\frac{1}{s} \right] \xrightarrow{\text{replace units}} 230.998 \text{ } [MHz]$$

The main observation here is that the time constant associated with the source node now limits the bandwidth, due the large Cgs (long channel). In Example 4-4, the bandwidth was limited by the time constant at the drain node.

4.5

$$I_D := 150 \text{ [}\mu\text{A]} : g_m := 0.75 \text{ [mS]} : R_{diode} := 50 \text{ [k}\Omega\text{]} : C_{diode} := 70 \text{ [fF]} : C_L := 150 \text{ [fF]} : R_L := 1 \text{ [k}\Omega\text{]} : L := 1 \text{ [um]} : V_{DD} := 2.5 \text{ [V]} : V_{SS} := -2.5 \text{ [V]} : R_G := 500 \text{ [}\Omega\text{]} :$$

(a) The gate overdrive, device width and bias current are computed as follows.

$$V_{OV} := \sqrt{\frac{2 \cdot I_D}{\mu_n C_{ox} \cdot \frac{W}{L}}} = 0.245 \text{ [V]}$$

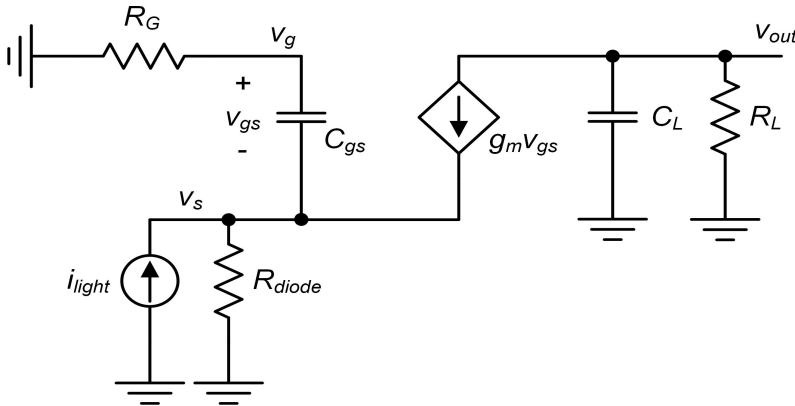
$$W := \frac{g_m^2}{2 \cdot I_D \cdot \mu_n C_{ox} \cdot \frac{1}{L}} = 0.000 \text{ [m]} \xrightarrow{\text{replace units}} 37.500 \text{ [um]}$$

$$V_S := -V_{Tn} - V_{OV} = -1.020 \text{ [V]}$$

$$I_{Rdiode} := \frac{V_{DD} - V_S}{R_{diode}} = 0.000 \text{ [A]} \xrightarrow{\text{replace units}} 70.406 \text{ [}\mu\text{A]}]$$

$$I_{BIAS} := I_{Rdiode} + I_D = 0.000 \text{ [A]} \xrightarrow{\text{replace units}} 220.406 \text{ [}\mu\text{A]}]$$

(b) The small-signal model can be constructed as follows. Note that the simplifications of Figure 4-10 cannot be applied, since the gate node is not an ac ground (due to R_G). We therefore need to go back to first principles.



(c) At low frequencies, we see $1/g_m$ looking into the controlled source from the node v_s (the presence of R_G is irrelevant). The transresistance is therefore found using the current divider at the source node, times R_L .

$$R_x := \frac{g_m}{g_m + \frac{1}{R_{diode}}} \cdot R_L = 0.974 \text{ [k}\Omega\text{]}$$

(d) Without R_G , we see $1/g_m$ looking into the controlled source from the node v_s , also at high

frequencies (the simplified model of Figure 4-13 applies). The bandwidth is estimated by simply computing the time constants at the drain and source terminals.

$$C_{gs} := \frac{2}{3} \cdot W \cdot L \cdot C_{ox} = 5.750 \cdot 10^{-14} \text{ [F]}$$

$$\tau_{so} := \left(g_m + \frac{1}{R_{diode}} \right)^{-1} \cdot C_{gs} = 7.468 \cdot 10^{-11} \text{ [s]}$$

$$\tau_{do} := 1 \cdot R_L \cdot C_L = 1.500 \cdot 10^{-10} \text{ [s]}$$

$$f_{3dB} := \frac{1}{2 \cdot 3.142} \cdot \frac{1}{\tau_{so} + \tau_{do}} = 7.083 \cdot 10^8 \left[\frac{1}{s} \right] \xrightarrow{\text{replace units}} 708.286 \text{ [MHz]}$$

(e) The time constant at the drain node stays as is. The time constant due to C_{gs} is now no longer obvious. We can go back to first principle analysis, apply a test voltage between gate and source and find the Thevenin resistance using vt/it. This was already done in P3.16, and we found that the resistance seen by C_{gs} is

$$R_{gso} := \frac{R_G + R_{diode}}{1 + g_m \cdot R_{diode}} = 1311.688 \text{ [\Omega]}$$

$$\tau_{so} := R_{gso} \cdot C_{gs} = 7.542 \cdot 10^{-11} \text{ [s]}$$

$$f_{3dB} := \frac{1}{2 \cdot 3.142} \cdot \frac{1}{\tau_{so} + \tau_{do}} = 7.059 \cdot 10^8 \left[\frac{1}{s} \right] \xrightarrow{\text{replace units}} 705.939 \text{ [MHz]}$$

We conclude that R_G has negligible impact on the circuit's bandwidth. More insight can be gained by approximating R_{gso} as

$$R_{gso} = \frac{R_G + R_{diode}}{1 + g_m \cdot R_{diode}} = \frac{R_G + R_{diode}}{g_m \cdot R_{diode}} = \frac{1}{g_m} \left(1 + \frac{R_G}{R_{diode}} \right)$$

Unless R_G is made comparable to R_{diode} , it will have little effect on the bandwidth (R_{gso} stays close to $1/g_m$, as in part e).

4.6

$$R_S := 'R_S'; R_D := 'R_D'; C_s := 'C_s'; C_d := 'C_d';$$

Apply a test voltage at vout and write KCL for the two nodes of the circuit:

$$\begin{aligned} Eq1 &:= 0 = v_s \cdot \left(\frac{1}{R_S} + s \cdot C_s + g_{mprime} \right) + \frac{v_s - v_t}{r_o} : \\ Eq2 &:= 0 = -i_t - g_{mprime} \cdot v_s + v_t \cdot s \cdot C_d + \frac{v_t - v_s}{r_o} : \\ collect(simplify(solve(\{Eq1, Eq2\}, [i_t, v_s])), s) \\ &\left[\left[i_t = \frac{v_t (1 + s^2 C_d r_o C_s R_S + (C_s R_S + C_d r_o + C_d r_o g_{mprime} R_S + C_d R_S) s)}{r_o + r_o s C_s R_S + r_o g_{mprime} R_S + R_S}, v_s \right. \right. \\ &\quad \left. \left. = \frac{R_S v_t}{r_o + r_o s C_s R_S + r_o g_{mprime} R_S + R_S} \right] \right] \end{aligned} \quad (1)$$

Using g'mRs >>1 and g'mro >>1, we have:

$$\begin{aligned} \frac{1}{Z_{in}} &= \frac{i_t}{v_t} = \frac{1 + r_o s^2 C_s R_S C_d + (C_s R_S + r_o g_{mprime} R_S C_d) s}{r_o s C_s R_S + r_o g_{mprime} R_S} : \\ \frac{1}{Z_{in}} &= \frac{1}{g_{mprime} r_o R_S} \frac{r_o C_s R_S C_d s^2 + (C_s R_S + r_o g_{mprime} R_S C_d) s + 1}{\left(\frac{s C_s}{g_{mprime}} s + 1 \right)} : \end{aligned}$$

Note that for s = 0 (DC), this result matches equation (4.22). Also, note that the numerator of the second term can be neglected for frequencies below fT.

$$\begin{aligned} \frac{1}{Z_{in}} &= \frac{r_o C_s R_S C_d s^2 + (C_s R_S + r_o g_{mprime} R_S C_d) s + 1}{g_{mprime} r_o R_S} : \\ \frac{1}{Z_{in}} &= \frac{1}{g_{mprime} r_o R_S} + \frac{C_s}{g_{mprime} r_o} s + s \cdot C_d \left(1 + \frac{C_s}{g_{mprime}} s \right) : \end{aligned} \quad (2)$$

Again, the last term inside the bracket is small unless we get close to the transistor's transit frequency. Hence, the final result matches (4.27).

$$\frac{1}{Z_{in}} = \frac{1}{g_{mprime} r_o R_S} + \frac{C_s}{g_{mprime} r_o} s + s \cdot C_d :$$

Note: If you want to save time in this derivation, do not include Cd in the original KCL equations and simply add it to the final result as a parallel combination. This can save time when doing this analysis manually.

4.7

$$R_{L1} := 1 \llbracket k\Omega \rrbracket : R_{L2} := 1 \llbracket M\Omega \rrbracket : W := 500 \llbracket \mu m \rrbracket : L := 1 \llbracket \mu m \rrbracket : V_{DD} := 5 \llbracket V \rrbracket : v_{OUTH} := 4.5 \llbracket V \rrbracket : \\ v_{OUTL} := 2 \llbracket V \rrbracket :$$

(a) We begin by computing the drain currents corresponding to the two given output voltage levels and for both resistor values.

$$I_{DH1} := \frac{V_{DD} - v_{OUTH}}{R_{L1}} = 0.001 \llbracket A \rrbracket \xrightarrow{\text{replace units}} 500.000 \llbracket \mu A \rrbracket$$

$$I_{DL1} := \frac{V_{DD} - v_{OUTL}}{R_{L1}} = \frac{3}{1000} \llbracket A \rrbracket \xrightarrow{\text{replace units}} 3000 \llbracket \mu A \rrbracket$$

$$I_{DH2} := \frac{V_{DD} - v_{OUTH}}{R_{L2}} = 5.000 \cdot 10^{-7} \llbracket A \rrbracket \xrightarrow{\text{replace units}} 0.500 \llbracket \mu A \rrbracket$$

$$I_{DL2} := \frac{V_{DD} - v_{OUTL}}{R_{L2}} = \frac{3}{1000000} \llbracket A \rrbracket \xrightarrow{\text{replace units}} 3 \llbracket \mu A \rrbracket$$

For the circuit in Figure P4.7(a), the corresponding input voltages are:

$$v_{INH1a} := v_{OUTH} + V_{T0p} - \sqrt{\frac{2 \cdot I_{DH1}}{\mu n C_{ox} \cdot \frac{W}{L}}} = 3.800 \llbracket V \rrbracket$$

$$v_{INL1a} := v_{OUTL} + V_{T0p} - \sqrt{\frac{2 \cdot I_{DL1}}{\mu n C_{ox} \cdot \frac{W}{L}}} = 1.010 \llbracket V \rrbracket$$

$$v_{INH2a} := v_{OUTH} + V_{T0p} - \sqrt{\frac{2 \cdot I_{DH2}}{\mu n C_{ox} \cdot \frac{W}{L}}} = 3.994 \llbracket V \rrbracket$$

$$v_{INL2a} := v_{OUTL} + V_{T0p} - \sqrt{\frac{2 \cdot I_{DL2}}{\mu n C_{ox} \cdot \frac{W}{L}}} = 1.485 \llbracket V \rrbracket$$

In the circuit in Figure P4.7(b), the threshold voltages vary with the output. We can compute the thresholds for the given levels as follows. Next, we compute the input voltages in the same way as before.

$$V_{TpH} := V_{T0p} - \gamma_p \cdot \left(\sqrt{2 \cdot \phi_f + V_{DD} - v_{OUTH}} - \sqrt{2 \cdot \phi_f} \right) = -0.647 \llbracket V \rrbracket$$

$$V_{TpL} := V_{T0p} - \gamma_p \cdot \left(\sqrt{2 \cdot \phi_f + V_{DD} - v_{OUTL}} - \sqrt{2 \cdot \phi_f} \right) = -1.133 \llbracket V \rrbracket$$

$$v_{INH1b} := v_{OUTH} + V_{TpH} - \sqrt{\frac{2 \cdot I_{DH1}}{\mu n C_{ox} \cdot \frac{W}{L}}} = 3.653 \llbracket V \rrbracket$$

$$v_{INL1b} := v_{OUTL} + V_{TpL} - \sqrt{\frac{2 \cdot I_{DL1}}{\mu n C_{ox} \cdot \frac{W}{L}}} = 0.377 \text{ [V]}$$

$$v_{INH2b} := v_{OUTH} + V_{TpH} - \sqrt{\frac{2 \cdot I_{DH2}}{\mu n C_{ox} \cdot \frac{W}{L}}} = 3.846 \text{ [V]}$$

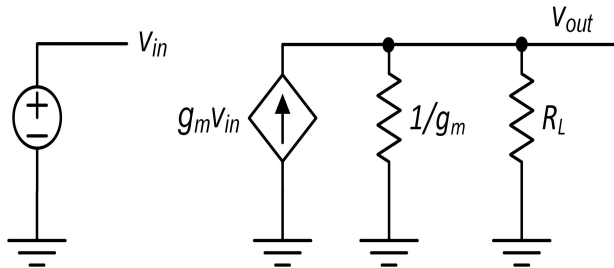
$$v_{INL2b} := v_{OUTL} + V_{TpL} - \sqrt{\frac{2 \cdot I_{DL2}}{\mu n C_{ox} \cdot \frac{W}{L}}} = 0.852 \text{ [V]}$$

Result Summary

	Figure P4-7(a)	Figure P4-7(b)
$R_L=1k, v_{OUT}=4.5V$	$v_{INH1a} = 3.800 \text{ [V]}$	$v_{INH1b} = 3.653 \text{ [V]}$
$R_L=1k, v_{OUT}=2V$	$v_{INL1a} = 1.010 \text{ [V]}$	$v_{INL1b} = 0.377 \text{ [V]}$
$R_L=1M, v_{OUT}=4.5V$	$v_{INH2a} = 3.994 \text{ [V]}$	$v_{INH2b} = 3.846 \text{ [V]}$
$R_L=1M, v_{OUT}=2V$	$v_{INL2a} = 1.485 \text{ [V]}$	$v_{INL2b} = 0.852 \text{ [V]}$

We note that the input swing is the smallest (and close to the output swing of 2.5V) for the circuit of Figure P4-7(a) with $R_L=1M$. This is because the gate overdrive of the MOSFET stays nearly constant due to the small current excursion. The polar opposite is the configuration of Figure P4-7(b) with $R_L=1k$. Here, the current excursion is large, and in addition the threshold voltage change requires extra input swing. The input swing is here $v_{INH1b} - v_{INL1b} = 3.275 \text{ [V]}$, much larger than 2.5V. Also, it is interesting to note that the input voltage swing is much larger than 2.5V for the case $R_L=1M$ and with backgate effect.

(b) The small-signal model is similar to Figure 4-17(c), but here we neglect r_o and g_{mb} is irrelevant, since $v_{bs}=0$.



The small-signal voltage gain can be expressed as follows:

$$A_v = \frac{v_{out}}{v_{in}} = g_m \cdot \frac{R_L \cdot \frac{1}{g_m}}{R_L + \frac{1}{g_m}} = \frac{1}{1 + \frac{1}{g_m R_L}}$$

$$g_m = \sqrt{2 I_D \cdot \mu n C_{ox} \cdot \frac{W}{L}} = \sqrt{2 \frac{V_{DD} - V_{OUT}}{R_L} \cdot \mu n C_{ox} \cdot \frac{W}{L}}$$

$$A_v = \frac{1}{1 + \frac{1}{\sqrt{2 \cdot R_L \cdot (V_{DD} - V_{OUT}) \cdot \mu n C_{ox} \cdot \frac{W}{L}}}}$$

For the given numbers, we have:

$$A_{vL} := \frac{1}{1 + \frac{1}{\sqrt{2 \cdot R_{LI} \cdot (V_{DD} - v_{OUTL}) \cdot \mu n C_{ox} \cdot \frac{W}{L}}}} = 0.925 \cdot 1$$

$$A_{vH} := \frac{1}{1 + \frac{1}{\sqrt{2 \cdot R_{LI} \cdot (V_{DD} - v_{OUTH}) \cdot \mu n C_{ox} \cdot \frac{W}{L}}}} = 0.833 \cdot 1$$

4.8

$$R_L := 1 \llbracket k\Omega \rrbracket : W := 40 : V_{DD} := 2 \llbracket V \rrbracket : V_{OUT} := 'V_{OUT}':$$

(a) The MOSFET operates in the triode region when $V_{DS} \leq V_{GS} - V_{Tn}$. Note that $V_{DS} = V_{DD} - V_{OUT}$ and $V_{GS} - V_{Tn} = V_{IN} - V_{OUT} - V_{Tn}$. Therefore, at the onset of triode operation, we have

$$V_{IN} := V_{DD} + V_{T0n} = 2.500 \llbracket V \rrbracket$$

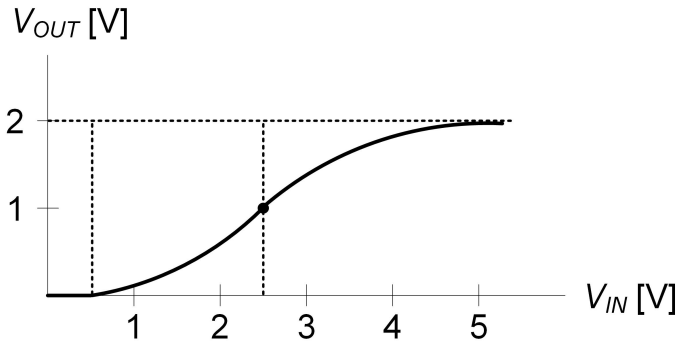
To find the corresponding output voltage, we can write the following equation and solve the resulting quadratic equation

$$V_{OUT} = V_{IN} - V_{T0n} - \sqrt{\frac{2 \cdot I_D}{\mu n C_{ox} \cdot W}} :$$

$$V_{OUT} \cdot \llbracket V \rrbracket = V_{IN} - V_{T0n} - \sqrt{\frac{\frac{2 \cdot V_{OUT} \cdot \llbracket V \rrbracket}{R_L}}{\mu n C_{ox} \cdot W}} :$$

$$\text{solve} \left(V_{OUT} \llbracket V \rrbracket = V_{IN} - V_{T0n} - \sqrt{\frac{\frac{2 \cdot V_{OUT} \llbracket V \rrbracket}{R_L}}{\mu n C_{ox} \cdot W}}, V_{OUT} \right) = 1.000$$

(b) For $V_{IN} = 0 \dots V_{Tn}$, the transistor is off and V_{OUT} is zero. For larger V_{IN} , the transistor is on and operates in the saturation until we reach $V_{IN} = 2.5V$, $V_{OUT} = 1V$ (computed above). It is important to note that the first derivative of the curve must be monotonically increasing in this range (since g_m is increasing with V_{IN}). For the remaining range from 2.5V to 5V, the transistor is in the triode region and the output voltage must asymptotically approach $V_{DD} = 2V$.



4.9

$$R_S := 'R_S'; R_L := 'R_L'; C_{gs} := 'C_{gs}'; C_{sb} := 'C_{sb}'; C_{gd} := 'C_{gd}'; g_m := 'g_m';$$

Writing KCL at nodes v_{in} and v_{out} gives

$$Eq1 := 0 = -\frac{v_s}{R_S} + v_{in} \cdot \left(\frac{1}{R_S} + s \cdot C_{gd} \right) + (v_{in} - v_{out}) \cdot s \cdot C_{gs} :$$

$$Eq2 := 0 = -g_m \cdot v_{in} + v_{out} \cdot \left(\frac{1}{R_{tot}} + s \cdot C_{tot} \right) + (v_{out} - v_{in}) \cdot s \cdot C_{gs} :$$

$$collect(solve(\{Eq1, Eq2\}, [v_{out}, v_{in}]), s)$$

$$\left[\left[v_{out} = (v_s R_{tot} (s C_{gs} + g_m)) / (1 + (C_{tot} R_{tot} C_{gd} R_S + C_{tot} R_{tot} C_{gs} R_S + C_{gs} R_{tot} C_{gd} R_S) s^2 + (-C_{gs} R_S g_m R_{tot} + C_{gd} R_S + C_{gs} R_S + C_{tot} R_{tot} + C_{gs} R_{tot}) s), v_{in} = (v_s (1 + (C_{tot} R_{tot} + C_{gs} R_{tot}) s)) / (1 + (C_{tot} R_{tot} C_{gd} R_S + C_{tot} R_{tot} C_{gs} R_S + C_{gs} R_{tot} C_{gd} R_S) s^2 + (-C_{gs} R_S g_m R_{tot} + C_{gd} R_S + C_{gs} R_S + C_{tot} R_{tot} + C_{gs} R_{tot}) s) \right] \right] \quad (3)$$

Therefore, we have

$$\frac{v_{out}}{v_s} = \frac{R_{tot} (g_m + s C_{gs})}{1 + b_1 s + b_2 s^2}$$

The numerator can be rewritten as follows, and therefore matches (4.36) and (4.39):

$$N = g_m \cdot R_{tot} \left(1 + s \frac{C_{gs}}{g_m} \right) = \frac{g_m}{g'_m + \frac{1}{R_L}} \left(1 + s \frac{C_{gs}}{g_m} \right) = A'_v \left(1 + s \frac{C_{gs}}{g_m} \right) :$$

Finally, we can also confirm (4.37) and (4.38) from the above result:

$$b_2 = R_S R_{tot} (C_{gd} C_{gs} + C_{gs} C_{tot} + C_{gd} C_{tot}) :$$

$$b_1 = C_{tot} R_{tot} + C_{gs} R_{tot} + C_{gd} R_S - C_{gs} R_S R_{tot} g_m + C_{gs} R_S = C_{tot} R_{tot} + C_{gs} R_{tot} + C_{gd} R_S + C_{gs} R_S (1 - A'_v) :$$

4.10

The circuit has three capacitors (C_{gd} , C_{tot} and C_{gs}), and hence there are three open-circuit time constants. Finding the time constants associated with C_{gd} and C_{tot} is straightforward.

$$\tau_{gdo} := R_S \cdot C_{gd} :$$

$$\tau_{toto} := R_{tot} \cdot C_{tot} :$$

The time constant associated with C_{gs} can be found from first principles by writing KCL/KVL for the circuit of Figure 4-20. We remove all capacitors and applying a test current source across the terminals of C_{gs} (similar to Figure 3-19).

$$Eq1 := v_t = v_{in} + R_{tot}(i_t - g_m \cdot v_{in}) :$$

$$Eq2 := v_{in} = i_t \cdot R_S :$$

$$solve(\{Eq1, Eq2\}, [v_t, v_{in}])$$

$$[[v_t = i_t R_S + R_{tot}(i_t - g_m i_t R_S), v_{in} = i_t R_S]] \quad (4)$$

Therefore, we find

$$R_{Tgs} = \frac{v_t}{i_t} = R_S + R_{tot}(1 - g_m R_S) = R_{tot} + R_S(1 - g_m R_{tot}) = R_{tot} + R_S(1 - A'_v) :$$

$$\tau_{gso} := R_{tot} C_{gs} + R_S C_{gs}(1 - A'_v) :$$

This confirms (4.37)

$$b_1 = \tau_{gdo} + \tau_{toto} + \tau_{gso} = R_S \cdot C_{gd} + R_{tot} \cdot C_{tot} + R_{tot} C_{gs} + R_S C_{gs}(1 - A'_v) :$$

4.11

$$C_{gs} := 200 \text{ [fF]} : gmro := 50 :$$

We solve this problem by computing the voltage gain of the circuit (which will be very close to 1), and then apply the result from Figure 4-21.

$$A'_v := \frac{1.}{1 + \frac{1}{gmro}} = 0.980$$

$$C_{in, gs} := C_{gs} \cdot (1 - A'_v) = 3.922 \text{ [fF]}$$

4.12

$$V_{OUT} := 1 \llbracket V \rrbracket : V_{OV} := 0.4 \llbracket V \rrbracket : f_T := 1 \llbracket GHz \rrbracket : R_L := 1 \llbracket k\Omega \rrbracket :$$

(a) From the given information, we can calculate I_D , which defines g_m via VOV

$$I_D := \frac{V_{OUT}}{R_L} = \frac{1}{1000} \llbracket A \rrbracket \xrightarrow{\text{replace units}} 1000 \llbracket \mu A \rrbracket$$

$$g_m := \frac{2 \cdot I_D}{V_{OV}} = 0.005 \llbracket S \rrbracket \xrightarrow{\text{replace units}} 5.000 \llbracket mS \rrbracket$$

$$A_v := \frac{1}{1 + \frac{1}{g_m \cdot R_L}} = 0.833$$

(b) With $R_S=0$, equation (4.36) becomes

$$\frac{v_{out}}{v_{in}} = A_v \cdot \frac{1 + s \cdot \frac{C_{gs}}{g_m}}{1 + s \cdot R_{tot}(C_{gs} + C_L)}$$

$$R_{tot} := \left(g_m + \frac{1}{R_L} \right)^{-1} = 166.667 \llbracket \frac{1}{S} \rrbracket \xrightarrow{\text{replace units}} 166.667 \llbracket \Omega \rrbracket$$

$$C_{gs} := \frac{g_m}{2 \cdot 3.142 \cdot f_T} = 7.957 \cdot 10^{-13} \llbracket F \rrbracket$$

The frequency response is flat when the pole and zero cancel, i.e.

$$\frac{C_{gs}}{g_m} = R_{tot}(C_{gs} + C_L) :$$

$$\frac{C_{gs}}{g_m} = R_{tot}(C_{gs} + C_L) :$$

$$C_L = C_{gs} \cdot \frac{\left(\frac{1}{g_m} - R_{tot} \right)}{R_{tot}} = C_{gs} \cdot \left(\frac{1}{g_m \cdot R_{tot}} - 1 \right) :$$

$$C_L := C_{gs} \cdot \left(\frac{1}{A_v} - 1 \right) = 1.591 \cdot 10^{-13} \llbracket F \rrbracket$$

As an aside, note that

$$\frac{C_{gs}}{C_{gs} + C_L} = 0.833$$

This means that the gain at very high frequencies, given by the capacitive divider, is the same as the low

frequency gain. This explains intuitively why the frequency response must be flat.

(c) With $CL=0$, the high-frequency voltage gain must be unity based on the intuition established above. We can use the transfer function to confirm this formally.

$$\frac{v_{out}}{v_{in}} = A_v \cdot \frac{1 + s \cdot \frac{C_{gs}}{g_m}}{1 + s \cdot R_{tot} C_{gs}}$$

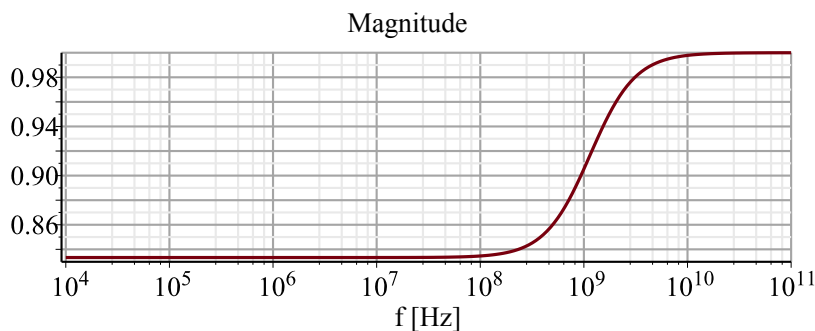
For $s \rightarrow \infty$, the transfer function approaches

$$A_v \cdot \frac{\frac{C_{gs}}{g_m}}{R_{tot} C_{gs}} = g_m R_{tot} \frac{g_m}{R_{tot}} = 1 :$$

The magnitude of the transfer function looks as follows.

$$H := s \rightarrow A_v \cdot \frac{1 + s \cdot \frac{C_{gs}}{g_m} \llbracket Hz \rrbracket}{1 + s \cdot R_{tot} C_{gs} \llbracket Hz \rrbracket} :$$

`semilogplot(|H(j·2·π·f)|, f=104 .. 1011, gridlines, title="Magnitude", labels=["f [Hz]", ""])`



As expected, the voltage gain changes from 0.833 at low frequencies to 1 at high frequencies. The zero in the transfer function occurs before the pole.

4.13

$$R_S := 5 \llbracket k\Omega \rrbracket : R_L := 5 \llbracket k\Omega \rrbracket : g_m := 2 \llbracket mS \rrbracket : g_{mb} := 0.417 \llbracket mS \rrbracket : C_{gs} := 203 \llbracket fF \rrbracket : C_{gd} := 50 \llbracket fF \rrbracket : C_{sb} := 54.3 \llbracket fF \rrbracket :$$

Note: The first edition of the book contains a typo in Example 4-6 (see errata). The correct value of RS = 5k (instead of 10k) is used in this solution.

$$R_{tot} := \left(g_m + g_{mb} + \frac{1}{R_L} \right)^{-1} = 382.117 \llbracket \frac{1}{S} \rrbracket \xrightarrow{\text{replace units}} 382.117 \llbracket \Omega \rrbracket$$

$$A'_v := g_m \cdot R_{tot} = 0.764$$

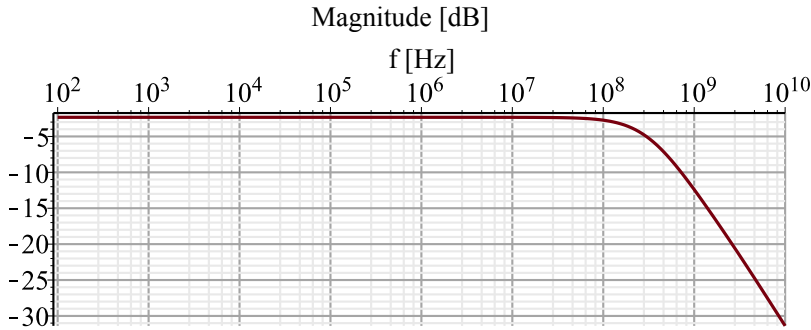
$$b_1 := R_S \cdot C_{gd} + R_{tot} \cdot C_{sb} + R_{tot} \cdot C_{gs} + R_S \cdot C_{gs} \cdot (1 - A'_v) = 5.876 \cdot 10^{-10} \llbracket s \rrbracket$$

$$b_2 := R_S \cdot R_{tot} \cdot (C_{gd} \cdot C_{gs} + C_{gs} \cdot C_{sb} + C_{gd} \cdot C_{sb}) = 4.564 \cdot 10^{-20} \llbracket s^2 \rrbracket$$

$$H13 := s \rightarrow A'_v \cdot \frac{1 + s \cdot \frac{C_{gs}}{g_m} \llbracket Hz \rrbracket}{1 + b_1 \cdot s \llbracket Hz \rrbracket + b_2 \cdot s^2 \llbracket Hz \rrbracket \llbracket Hz \rrbracket} :$$

$$Mag := x \rightarrow 20 \cdot \log_{10}(|x|):$$

$$\text{semilogplot}(Mag(H13(j \cdot 2 \cdot \pi \cdot f)), f = 10^2 .. 10^{10}, \text{gridlines}, \text{title} = \text{"Magnitude [dB]"}, \text{labels} = [f \llbracket Hz \rrbracket, ""])$$



$$f_{3dB} := fsolve \left(\left| \frac{1 + \frac{C_{gs}}{g_m} \cdot (j \cdot 2 \cdot \pi \cdot f \llbracket MHz \rrbracket)}{1 + b_1 \cdot (j \cdot 2 \cdot \pi \cdot f \llbracket MHz \rrbracket) + b_2 \cdot (j \cdot 2 \cdot \pi \cdot f \llbracket MHz \rrbracket)^2} \right| = \frac{1}{\sqrt{2}}, f, \text{real} \right) = 323.514$$

For reference, we compare this value to the OCT estimate computed in Example 4-6:

$$f_{3dB, OCT} := \frac{1}{2 \cdot 3.14 \cdot b_1} = 2.710 \cdot 10^8 \llbracket \frac{1}{s} \rrbracket \xrightarrow{\text{replace units}} 270.983 \llbracket MHz \rrbracket$$

As expected, the exact bandwidth is somewhat larger than the OCT estimate. Interestingly, the exact bandwidth is close to the value predicted by the simple expression of (4.41) (see Example 4-6).

4.14

$$R_{tot} := 'R_{tot}'; C_{gs} := 'C_{gs}'; g_m := 'g_m';$$

In Figure 4-21, the voltage gain from v_{in} to v_{out} follows from (4.36) with $R_S=0$, $C_{sb}\sim 0$ (neglected relative to C_{gs}) and $C_L\sim 0$ (neglected relative to C_{gs}).

$$\frac{v_{out}}{v_{in}} = A'_v \cdot \frac{1 + s \cdot \frac{C_{gs}}{g_m}}{1 + s \cdot R_{tot} \cdot C_{gs}} :$$

Since $R_{tot} < 1/g_m$, both the pole and zero in this expression occur near or beyond f_T of the transistor. Hence, it is valid to approximate the gain from v_{in} to v_{out} as constant, as argued in Section 4-3-3.

$$\frac{v_{out}}{v_{in}} \cong A'_v :$$

4.15

$$R_S := 'R_S'; C_{gs} := 'C_{gs}'; C_{gd} := 'C_{gd}'; g_m := 'g_m';$$

We consider the circuit of Figure 4-23, neglecting C_{sb} and approximating $g'_m = g_m$. KCL gives:

$$Eq1 := 0 = v_{in} \cdot \left(\frac{1}{R_S} + s \cdot C_{gd} \right) + (v_{in} - v_t) \cdot s \cdot C_{gs} :$$

$$Eq2 := 0 = -g_m \cdot v_{in} + v_t \cdot g_m + (v_t - v_{in}) \cdot s \cdot C_{gs} - i_t :$$

$$collect(simplify(solve(\{Eq1, Eq2\}, [v_t, v_{in}]), s))$$

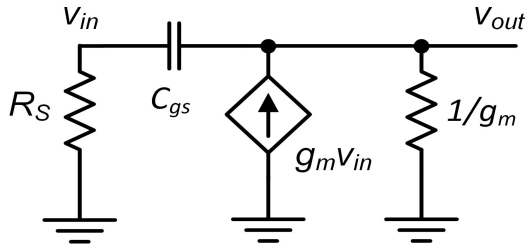
$$\left[\left[v_t = \frac{((C_{gs} R_S + C_{gd} R_S) s + 1) i_t}{(s C_{gs} + g_m) (1 + s C_{gd} R_S)}, v_{in} = \frac{s C_{gs} R_S i_t}{s^2 C_{gs} C_{gd} R_S + (C_{gs} + g_m C_{gd} R_S) s + g_m} \right] \right] \quad (5)$$

$$Z_{out} = \frac{v_t}{i_t} = \frac{1}{g_m} \cdot \frac{1 + s \cdot R_S \cdot (C_{gs} + C_{gd})}{\left(1 + \frac{s \cdot C_{gs}}{g_m} \right) \cdot (1 + s \cdot R_S \cdot C_{gd})} :$$

This confirms the result discussed in section 4-4-3. With the given assumptions, the output impedance has a zero that occurs first, followed by two poles.

4.16

(a) The small-signal model is shown below.



We apply a test voltage source at the output and write KCL at the two nodes of the circuit.

$$Eq1 := 0 = \frac{v_{in}}{R_S} + (v_{in} - v_t) \cdot s \cdot C_{gs} :$$

$$Eq2 := 0 = -g_m \cdot v_{in} + v_t \cdot g_m + (v_t - v_{in}) \cdot s \cdot C_{gs} - i_t :$$

$$collect(simplify(solve(\{Eq1, Eq2\}, [v_t, v_{in}]), s))$$

$$\left[\left[v_t = \frac{(1 + s C_{gs} R_S) i_t}{s C_{gs} + g_m}, v_{in} = \frac{s C_{gs} R_S i_t}{s C_{gs} + g_m} \right] \right] \quad (6)$$

$$Z_{out} = \frac{v_t}{i_t} = \frac{1}{g_m} \cdot \frac{1 + s \cdot R_S \cdot C_{gs}}{1 + \frac{s \cdot C_{gs}}{g_m}} :$$

(b) The impedance of the circuit in Figure P4-16 is:

$$L := 'L':$$

$$Z_{RL} = \frac{R_2 \cdot (R_1 + s \cdot L)}{R_2 + R_1 + s \cdot L} = \frac{R_2 \cdot R_1}{R_2 + R_1} \cdot \frac{1 + s \cdot \frac{L}{R_1}}{1 + s \cdot \frac{L}{R_2 + R_1}} :$$

Since $R_2 + R_1 > R_1$, this expression can model Z_{out} if and only if $g_m > 1/R_S$, i.e. $1/g_m < R_S$. We find expressions for L , R_1 and R_2 via coefficient comparison.

$$\frac{1}{g_m} = \frac{R_2 \cdot R_1}{R_2 + R_1} :$$

$$C_{gs} \cdot \frac{R_2 \cdot R_1}{R_2 + R_1} = \frac{L}{R_2 + R_1} :$$

$$L = C_{gs} \cdot R_2 \cdot R_1 :$$

$$\frac{L}{R_1} = C_{gs} \cdot R_2 = R_S \cdot C_{gs} :$$

$$R_2 = R_S :$$

$$g_m = \frac{1}{R_S} + \frac{1}{R_1} :$$

$$R_1 = \frac{1}{g_m - \frac{1}{R_S}} :$$

(c) We now compute L, R1 and R2 for the given parameters.

$$I_B := 1 \llbracket mA \rrbracket : V_S := 1 \llbracket V \rrbracket : R_S := 1 \llbracket M\Omega \rrbracket : W := 20 \llbracket um \rrbracket : L := 1 \llbracket um \rrbracket :$$

$$C_{gs} := \frac{2}{3} \cdot W \cdot L \cdot C_{ox} = 3.067 \cdot 10^{-14} \llbracket F \rrbracket$$

$$g_m := \sqrt{2 \cdot I_B \cdot \mu_n C_{ox} \cdot \frac{W}{L}} = 0.001 \llbracket S \rrbracket \xrightarrow{\text{replace units}} 1.414 \llbracket mS \rrbracket \qquad \frac{1}{g_m} = 707.107 \llbracket \Omega \rrbracket$$

$$R_2 := R_S = \llbracket M\Omega \rrbracket$$

$$R_1 := \frac{1}{g_m - \frac{1}{R_S}} = 707.607 \llbracket \Omega \rrbracket$$

$$L := C_{gs} \cdot R_2 \cdot R_1 = 0.000 \llbracket H \rrbracket \xrightarrow{\text{replace units}} 21.700 \llbracket uH \rrbracket$$